Sessional 1 (B.Tech VI Sem CSE)

BCE-C612 Formal Languages and Automata Theory

Duration: 1HR **Date: 19/2/25** **Time: 3:30 pm** **Max. Marks :20**

Section A

Note: Attempt any two questions. Each question carry 6 marks. **2 x 6 =12 M**

Questions	LO	BT
Q1. Define the term Finite Automata. Differentiate between NFA and DFA.	CO1	U
Q2. Write regular expression for the language over the alphabet {0,1}, with atmost two consecutive zeros.	CO1	U
Q3. Given the following Finite State Automaton (FSA) $M=(Q,\Sigma,\delta,q0,F)$ where: The transition function δ is defined as: $\delta(q0,a)=q1$, $\delta(q0,b)=q2$, $\delta(q1,a)=q0$ $\delta(q1,b)=q1$, $\delta(q2,a)=q2$, $\delta(q2,b)=q0$ .The initial state and final state is q0 . Is the string "ababb" accepted by the FSA.	CO3	E
Q4.Given the following Nondeterministic Finite Automaton (NFA) with null (ε) moves: Transition function δ $\delta(q0,a)=\{q0,q1\}$,$\delta(q0,b)=\{q0\}$,$\delta(q1,\varepsilon)=\{q2\}$,$\delta(q2,b)=\{q2\}$ $\delta(q2,a)=\{q1\}$. Initial state: q0,Accepting state: F={q2} Remove the null (ε) transitions from the given NFA.	CO2	K

Section B

Note: Attempt any one question from this section. **1x8 = 8 M**

Questions	LO	BT
Q1. Design a DFA equivalent for the given NFA; where q3 is the final state. [10]	CO1	U
Q2. Given the following Nondeterministic Finite Automaton (NFA) $M=(Q,\Sigma,\delta,q0,F)$ .The transition function $\delta\delta$ is defined as: $\delta(q0,a)=\{q0,q1\}$,$\delta(q0,b)=\{q0\}$,$\delta(q1,\varepsilon)=\{q2\}$,$\delta(q2,b)=\{q2\}$,$\delta(q2,a)=\{q1\}$.The initial state is q0 .The set of accepting states is F={q2} Is the string "ab" accepted by the NFA?	CO3	AP

Table for Q1:

Current State	Input 'a'	Input 'b'
→ q0	q0, q1	q0 , q2
q1	--	q3
q2	q0, q3	q1
q3	q2	-

Sessional II (B.Tech VI Sem CSE)

BCE-C612 Formal Languages and Automata Theory

Duration: 1HR **Date: 3/4/25** **Time: 3:30 pm** **Max. Marks :20**

Section A

2 x 6 =12 M

Note: Attempt any two questions. Each question carry 6 marks.

Questions	LO	BT
Q1. Define the term Push Down Automata	CO1	U
Q2. Construct PDA which accepts strings made of 0 and 1 and is a pallindrome.	CO3	E
Q3.For the given grammar do left most derivation for the string **aabaabbb;** $S \rightarrow ab \mid ba \mid aB \mid bA$; $A \rightarrow a \mid bAA$; $B \rightarrow b \mid aBB$.	CO2	k
Q4. Show the working of PDA by taking suitable example.	CO2	K

Section B

1x8 = 8 M

Note: Attempt any one question from this section.

Questions	LO	BT
Q1. Design CFG which accepts strings of type $a^i b^j$; i=j	CO3	E
Q2. Design a PDA for the string of type wcw^r , where w is made of a and b.	CO3	AP

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